AI Paper Trading Lab

Project overview, architecture, results so far, and what was fixed

October 2026 - three bots on paper accounts - no live track record yet

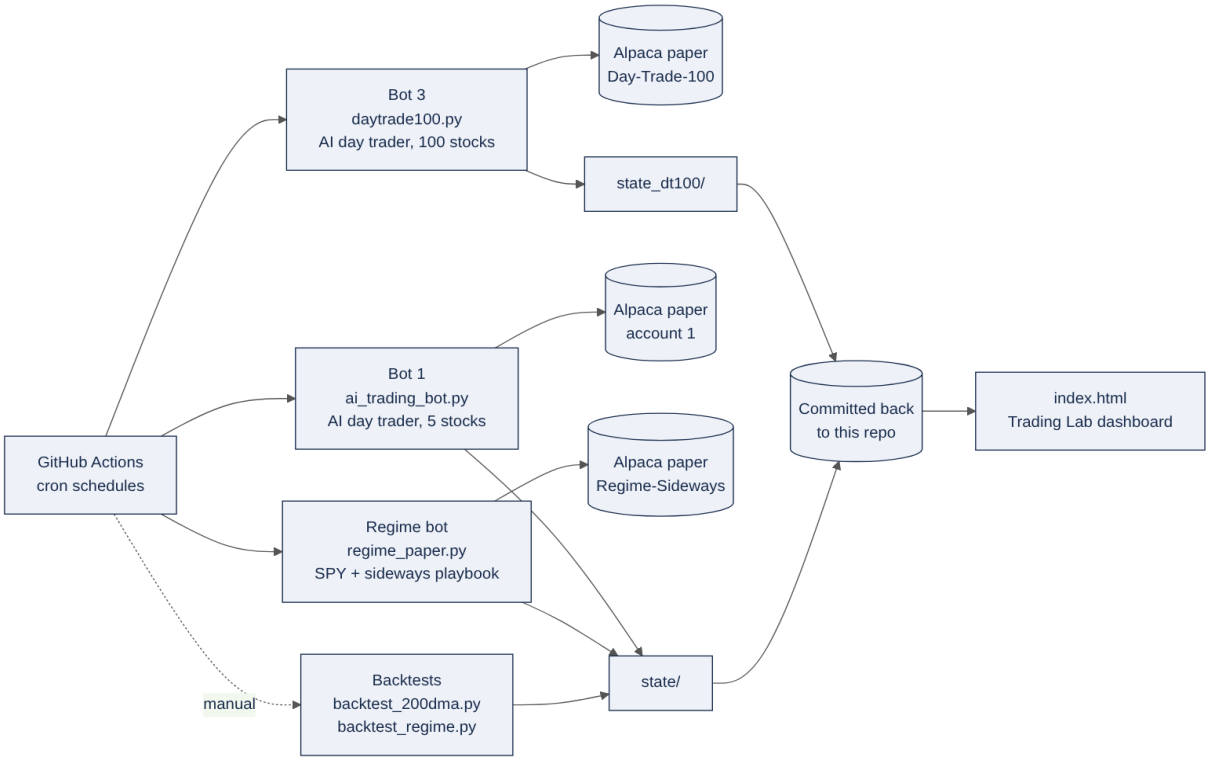
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1. What it is

A small lab of automated trading bots that run on a schedule in GitHub Actions, each on its own Alpaca paper account. Two are intraday AI bots: an XGBoost and LightGBM model scores stocks every 10 minutes from 54 features based on well-known traders' methods, and retrains every night, folding in the outcomes of its own trades. The third is a slower daily strategy that holds SPY unless the market is moving sideways. Every bot saves its memory back into the repository after each run, and a dashboard reads those files.

The goal is to find out, with honest measurement, whether any of these approaches beats simply holding SPY.

2. Architecture



Bot	What it does	Account	Runs
Bot 1 ai_trading_bot.py	AI day trader on a 5-stock watchlist, 15-minute bars. Signal: ML 50%, TradingView 35%, RSI 15%. Closes everything by 3:50 pm.	Paper account 1	Premarket, every 10 min, end of day, nightly retrain

Bot	What it does	Account	Runs
Bot 3 daytrade100.py	Same engine on the 100 most-traded S&P; 500 stocks; TradingView off; buys when the model clears its threshold and no safety veto fires.	Paper Day-Trade-100	Same as bot 1
Regime bot regime_paper.py	Holds SPY in bull and bear markets; when sideways, holds 10 oversold S&P; 500 stocks still above their 200-day average. Other playbooks run in shadow mode.	Paper Regime-Sideways ($25,000)	Daily, 3:45 pm
Backtests	200-day trend strategy and the regime strategy on S&P; 500 data since 2015, compared with SPY.	None	By hand

Risk rules on the AI bots: 2% of equity risked per trade, 12% position cap (6% in a bear market), 5 open positions at most, -2% stop loss, profit locks at +1.5% and +4%, a trailing stop from +6%, no buys near earnings or when VIX is 30 or more, and no overnight holds. Full details: docs/ARCHITECTURE.md.

3. Results so far

What	Result
Live paper trading (since October 18, 2026)	Bots 1 and 3 have placed no orders yet. The regime bot holds SPY, because the market is in a bull regime; it is level with SPY so far.
Model quality, bot 1	Walk-forward precision 31% to 37% when about 26% of training rows are real buy setups: a small edge over guessing. Accuracy about 51%.
Model quality, bot 3	Walk-forward precision 29% against a 24% base rate. Accuracy about 58%.
200-day trend backtest (6,245 trades, 2015-2026)	Average excess return over SPY +0.13% per trade; beat SPY on 41% of trades. No real edge.
Regime backtest v3 (2015-2026)	+15.9% a year against SPY's +13.7%; Sharpe 0.89 against 0.82; but a deeper worst drop (-36.2% against -33.7%). An upper bound, since 22% of past S&P; 500 stocks lack price data.

4. Problems found and fixed

Every fix below came from a failed run, a log or a backtest.

#	Problem	Fix
1	The bot was called self-learning, but the nightly retrain never used its own trades; it only printed a win rate.	Each closed trade now overwrites the label of the matching training row with the real outcome, counted 8 times.
2	The model, trade log and learning history were wiped on every Render redeploy.	State kept in a persistent folder, now committed back to the repository after every run.
3	The bot forgot its watchlist, peak prices and re-entry list overnight.	runtime_state.json carries them between runs.
4	Render hosting was not working and was not free.	Moved to free GitHub Actions schedules, one job per run (premarket, scan, end of day, retrain).
5	A neutral TradingView rating, the most common one, still took 35% of the score, so a buy was almost unreachable.	When TradingView is neutral or down, its weight moves to the model and the bar drops from 0.55 to 0.48.
6	A stock with no reversal pattern stayed on the re-entry watch list all day and could not be traded.	Re-entry waits at most 4 hours, then normal signals apply again.
7	The training label rewarded brief spikes that then fell sharply.	A row counts as a buy only if price rises 0.5% within 8 bars without first dropping 1%.
8	The buy threshold was a textbook 0.5 that was never checked.	Calibrated against precision at every retrain.
9	Nothing stopped buying during market panics.	VIX gate: half size at 25 or more, no new buys at 30 or more.
10	Bot 3's training weights overflowed to infinity on about 650,000 rows, so the model refused to train.	Time-decay rate changed so the oldest row weighs about 5%, safe up to millions of rows.
11	Bot 1's decay rate made nearly all 235,000 rows weigh close to zero, so the model said 50% for everything and never bought.	Same fix as bot 3: oldest row about 5%, newest 100%.
12	The final models stopped after 1 and 8 trees, so all five stocks scored exactly 0.45.	Final models use the tree count the walk-forward test found useful (100-400); a warning fires when scores barely vary.
13	Bot 3 made 0 trades in 13 scans: with TradingView off it needed about 72% confidence.	Buy when the model clears its threshold and no veto fires (RSI over 78, below the 50 EMA, top of Bollinger Band).
14	100 TradingView calls every 10 minutes would be rate-limited.	TradingView off for bot 3.
15	900 days of data for 100 stocks was too heavy for a free GitHub runner.	Bot 3 trains on 365 days.
16	No scheduled run fired after the workflows were added.	Workflow files re-saved so GitHub registered the schedules.
17	The first 200-day backtest used 15 hand-picked winners.	Rerun on all S&P; 500 stocks with every trade compared with SPY; it showed no edge.
18	The full regime strategy (v2) made +8.9% a year with a Sharpe of 0.45, well behind SPY.	v3 holds SPY in bull and bear markets and trades only the sideways playbook.
19	The Gemini model used for the news brief was retired in June 2026.	Switched to a current model; the brief is skipped when no key is set.

5. Honest assessment

Question	Answer today
Does it run?	Yes. Three bots on free schedules, each with its own paper account, saving their memory after every run.
Has it traded?	Not yet. The AI bots have placed no orders since moving to GitHub Actions, so there is no live result and nothing learned from real trades.
Does the model add value?	Weak and unproven: a few points of precision above the base rate. The README's 'about 80% accuracy' is not supported by the logs.
What drives the buy signals?	In bot 3, time of day and volatility rank highest, so the model may be learning when stocks move rather than which ones to buy.
Does anything beat SPY?	Only the regime strategy, modestly and only in a backtest that is an upper bound. The 200-day strategy does not.

6. Open problems

- No live track record yet; one month of paper results against SPY is the next real test.
- Cron times are set for daylight time and must move one hour later when daylight saving ends on November 1.
- 'hour' is listed twice among the model's features (55 entries, 54 unique).
- Bot 1's workflow uses a different lock from the other two, so their saves back to the repository can collide.
- No Gemini key is set, so the news ranking never runs and bot 1 always trades the same five stocks.
- The README still describes Render, Gemini, about 80% accuracy and a 3-position cap; the code uses 5.

7. Next steps

- Run all three bots for a month and compare each with SPY, as planned, before any real money.
- Fix the daylight-saving schedule before November 1.
- Test the model without time-of-day features; keep them only if precision holds.
- Measure backtested profit after costs for the ML signals, not just precision.
- Fix the duplicate feature and the workflow lock, and bring the README up to date.

Repo: github.com/shubhamkumarsingh91-lgtm/AI-Paper-Trading-Bot. Paper trading only. No employer or personal data is used in this project.